LU Decomposition

$$x_1 + x_2 - x_3 = 4$$
$$x_1 - 2x_2 + 3x_3 = -6$$
$$2x_1 + 3x_2 + x_3 = 7$$

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ -6 \\ 7 \end{bmatrix}$$

$$Ax = b$$

We can solve the system using LU Decomposition

Let $A = LU$ & substitute into $Ax = b$

Solve $LUx = b$ for x to solve the system

Let $Ux = y$

$Ly = b$ & $Ux = y$

First solve $Ly = b$ for y and then solve $Ux = y$ for x.

$$L = \begin{bmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ x & x & 1 \end{bmatrix} \quad U = \begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{bmatrix}$$

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$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \begin{matrix} R_1 \\ -1R_1 + R_2 \\ R_3 \end{matrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 2 & 3 & 1 \end{bmatrix} -2R_1 + R_3$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 1 & 3 \end{bmatrix} \frac{1}{3}R_2 + R_3$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & +13/3 \end{bmatrix} = U$$

① $-1R_1 + R_2$

$$\begin{array}{ccc} -1 & -1 & 1 \\ 1 & -2 & 3 \\ \hline 0 & -3 & 4 \end{array}$$

② $-2R_1 + R_3$

$$\begin{array}{ccc} -2 & -2 & 2 \\ 2 & 3 & 1 \\ \hline 0 & 1 & 3 \end{array}$$

③ $\frac{1}{3}R_2 + R_3$

$$\begin{array}{ccc} 0 & -1 & 4/3 \\ 0 & 1 & 3 \\ \hline 0 & 0 & -9/4 \end{array}$$

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First solve

$Ly = b$ using Foward substitution.

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/3 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \\ 7 \end{pmatrix}$$

$y_1 = 4$

$y_1 + y_2 = -6.$

$2y_1 + \frac{1}{3}y_2 + y_3 = 7$

$$y = \begin{pmatrix} 4 \\ -10 \\ -13/3 \end{pmatrix}.$$

$Ux = y$ (using Backward substitution)

$$\begin{pmatrix} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 13/3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ -10 \\ -13/3 \end{pmatrix}$$

$x_1 + x_2 - x_3 = 4$

$-3x_2 + 4x_3 = -10$

$13/3\, x_3 = -13/3.$

$$x = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

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